IEEE-IGNITE Hackathon 2026 | Problem SKIT002

MuleWatch

Demonstration Guide

AI-based detection of suspicious financial advertisements: posts that sell or rent bank accounts, ATM, debit and credit cards, current and savings accounts, and "full kits" used as money-mule accounts.

Event	IEEE-IGNITE Hackathon, 14–15 October 2026
Institute	Swami Keshvanand Institute of Technology (SKIT), Jaipur
Source code	https://github.com/vikashdhankar/mulewatch
Command center	http://localhost:8000 (presenter's laptop)
Public demo forum	http://103.155.85.18:8100 (Charcha)

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1. What you are demonstrating

MuleWatch is a real-time monitoring and investigation system. It crawls public platforms every 3 seconds, reads every new post (including the text inside images), scores how likely it is to be selling or renting banking credentials, links posts that come from the same seller network, and alerts the investigator within seconds. Each case comes with highlighted evidence and a downloadable PDF evidence report.

For the demo, the system watches **Charcha**, a public Reddit-style community forum we built and seeded with 50 realistic posts. Judges can open the forum on their own phones, log in, and post anything. MuleWatch catches mule-account ads as they are posted.

1.1 The two applications

Application	Purpose	Where it runs	Address
MuleWatch command center	Crawler, AI detection engine, investigation dashboard	Presenter's laptop	http://localhost:8000
Charcha forum	Public community forum used as the live target. Login, posting with text, links and images, comments, votes	CyberPanel server (port 8100)	http://103.155.85.18:8100
Charcha forum (backup)	Same forum running on the laptop if the server or internet fails	Presenter's laptop	http://localhost:8100

1.2 How each expected outcome is covered

Expected outcome (problem statement)	What MuleWatch does	Where to show it
Collect or process publicly available advertisement data	Live crawler every 3 s; reads schema.org JSON-LD that real forums publish; downloads attached images. Telegram public-channel crawler, Chrome extension and manual check page as extra inputs.	Monitored sources page; Live intercepts feed
Identify suspicious keywords, images, phone numbers, URLs and payment details	English, Hindi and Hinglish lexicon with obfuscation handling (acc0unt, a.t.m); phone numbers in any format; UPI IDs; Telegram/WhatsApp links; IFSC; card numbers (masked); crypto wallets; prices. OCR reads text inside images.	Case page: highlighted text, Identifiers, Text read from the image
Classify advertisements based on risk indicators	Hybrid score 0–100 from rule engine + multilingual text model + image cues (+ optional Claude). Levels Low / Medium / High / Critical; categories Mule Account Sale, Account-Holder Recruitment, Card Sale, KYC Fraud.	Risk badge and score meter; Why it was flagged; score breakdown
Detect repeated or coordinated advertisements	Near-duplicate text (MinHash), reposted or edited images (perceptual hash), shared phone/UPI/Telegram/wallet. Linked posts form seller rings and get a risk boost.	Seller rings panel; Seller network graph; Linked posts

Expected outcome (problem statement)	What MuleWatch does	Where to show it
Provide evidence / context supporting a suspicious classification	Highlighted words and identifiers, OCR text, per-component scores, linked posts with the reason for each link, SHA-256 evidence hash, PDF evidence report.	Case page; Evidence report (PDF)
Generate alerts and an investigation dashboard	Real-time alert toast with sound, optional Telegram push to phone, case management with verdicts, watchlist, takedown request to the platform.	Command center; Cases; Watchlist; Request takedown

2. Links, logins and ports

What	Value	Notes
Command center	http://localhost:8000	Open on the projector laptop. If port 8000 is busy, run.sh prints the port it used (e.g. 8001)
Public forum	http://103.155.85.18:8100	Judges open this on their phones
Backup forum	http://localhost:8100	Started with ./run.sh --local-forum
Backup forum for phones on the same Wi-Fi	http://<laptop-IP>:8100	Find the IP with `hostname -I`
API documentation	http://localhost:8000/docs	Interactive API (FastAPI)
GitHub repository	https://github.com/vikashdhankar/mulewatch	Code, README, tests
Forum demo login	demo / demo123	Presenter account
Forum judge logins	judge1, judge2, judge3 / demo123	Hand these to judges
Seeded forum users	<username> / charcha123	e.g. rohit_traders / charcha123
Moderation (takedown) token	skit-cybercell-2026	Must match on server and laptop .env
Ports	8000 (command center), 8100 (forum)	Open 8100 on the server firewall

> **Keep these three tabs open on the presenter laptop**
> 1. http://localhost:8000 (command center, sound on)
> 2. http://103.155.85.18:8100 (the forum, logged in as demo)
> 3. The GitHub repo, in case a judge asks about the code

3. One-time setup (do this before the event)

You need this only once. After that, use the activation checklist in section 4 every time you present.

3.1 Presenter laptop (Ubuntu)

Install system packages, download the code and start it once to install everything:

```
# system packages (Python virtualenv + OCR with Hindi)
sudo apt update
sudo apt install -y git python3-venv tesseract-ocr tesseract-ocr-hin

# get the code (if an old ~/Projects/mulewatch exists, move it aside first)
cd ~/Projects
[ -d mulewatch ] && mv mulewatch mulewatch_old
git clone https://github.com/vikashdhankar/mulewatch
cd mulewatch

# first start: creates .venv, installs packages, opens on :8000
./run.sh
```

The first start takes 1–2 minutes. When you see `MuleWatch command center: http://localhost:8000`, open that address. Stop it with **Ctrl+C**.

If GitHub asks for a password, the repository is private: enter your GitHub username and a personal access token (GitHub > Settings > Developer settings > Personal access tokens), or run `gh auth login` first.

3.2 Configure .env on the laptop

The first run creates `.env` from `.env.example`. Check these lines (open with `nano .env`):

```
FORUM_URL=http://103.155.85.18:8100
FORUM_MOD_TOKEN=skit-cybercell-2026
CRAWL_INTERVAL=3

# optional (section 3.4)
TELEGRAM_BOT_TOKEN=
TELEGRAM_CHAT_ID=

# optional AI analyst on the Check a post page
ANTHROPIC_API_KEY=
```

3.3 Forum on the CyberPanel server (103.155.85.18)

The forum runs as its own service on port 8100, so it does not touch the websites CyberPanel serves on ports 80 and 443.

a) Copy the forum to the server (run on the laptop)

```
cd ~/Projects
scp -r mulewatch/forum mulewatch/deploy root@103.155.85.18:/opt/charcha-src/
```

b) Install and start it (run on the server)

```
ssh root@103.155.85.18
sudo FORUM_MOD_TOKEN=skit-cybercell-2026 bash /opt/charcha-src/deploy/setup-forum-server.sh
```

The script installs Python packages into `/opt/charcha/venv`, seeds the 50 posts, creates a systemd service called `charcha`, and tries to open port 8100 in firewalld, ufw or CSF. It ends with a line starting with **Charcha is running**.

c) Open port 8100

- CyberPanel: **Security > Firewall** > add rule: name `charcha`, protocol TCP, port **8100** > Add.
- If your hosting provider has its own firewall or security group, allow TCP 8100 there too.

d) Verify from the laptop

```
curl -I http://103.155.85.18:8100        # expect: HTTP/1.1 200 OK
```

Then open http://103.155.85.18:8100 on your phone using mobile data (not the campus Wi-Fi), to be sure it is reachable from outside.

3.4 Optional: alerts on your phone through Telegram

1. In Telegram, open **@BotFather**, send `/newbot`, choose a name and username, and copy the token it gives you.
2. Open your new bot and send it any message (for example: hi).
3. In a browser open `https://api.telegram.org/bot<TOKEN>/getUpdates` and copy the number after `"chat":{"id":`.
4. Put both values in `.env` as `TELEGRAM_BOT_TOKEN` and `TELEGRAM_CHAT_ID`, then restart `./run.sh`.
5. The Monitored sources page now shows **Telegram push is on**. Critical alerts arrive on your phone. The first crawl of a source does not push old posts.

3.5 Optional: Chrome extension

1. Open `chrome://extensions` and turn on **Developer mode**.
2. Click **Load unpacked** and choose the folder `~/Projects/mulewatch/extension`.
3. On any website, select text or right-click an image and choose **Check selected text with MuleWatch**. The verdict appears as a notification and a badge, and the case is saved in the command center.

4. Activation checklist (30 minutes before presenting)

Follow these in order. Each command is safe to repeat.

4.1 Reset the forum to the original 50 posts (server)

This removes test posts and takedown flags from your rehearsal.

```
ssh root@103.155.85.18
sudo RESEED=1 FORUM_MOD_TOKEN=skit-cybercell-2026 bash /opt/charcha-src/deploy/setup-forum-server.sh
```

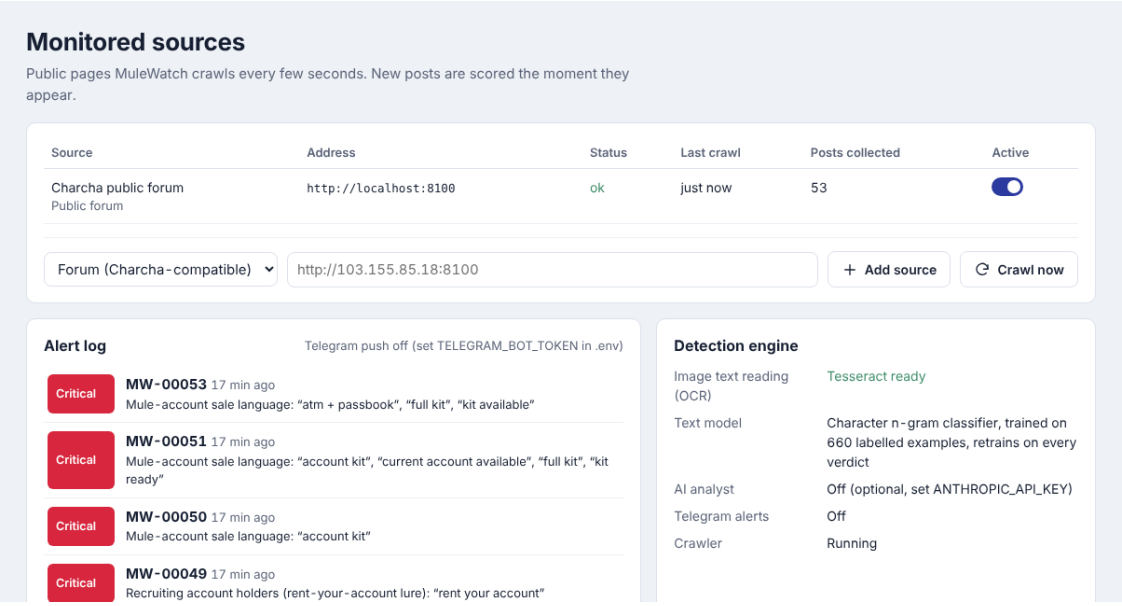
4.2 Start the command center with a clean slate (laptop)

```
cd ~/Projects/mulewatch
git pull                 # get the latest fixes
./run.sh --reset         # clears old cases, crawls the forum again
```

Leave this terminal open. Within about a minute the command center shows the full forum scanned.

4.3 Verify before judges arrive

Check	Expected	If not
Top bar of http://localhost:8000	Green dot, Monitoring 1 source, last crawl just now	See troubleshooting: source unreachable
Command center numbers	50 posts scanned, 16 critical, 4 seller rings	Wait one minute; if posts scanned is above 50, reset the forum (4.1)
Monitored sources > Detection engine	Image text reading: Tesseract ready	`sudo apt install tesseract-ocr tesseract-ocr-hin`, restart
Sound button (top right)	Click it so it reads Sound on. You hear a short chime.	Unmute the laptop; browsers need one click before playing sound
Projector	Browser zoom 110–125% so the back row can read	Ctrl + plus
Judges' phones	http://103.155.85.18:8100 opens; logged in as judge1, judge2	Use the backup plan in section 7



Monitored sources page: crawler status, alert log and detection engine status

Do a 30-second rehearsal

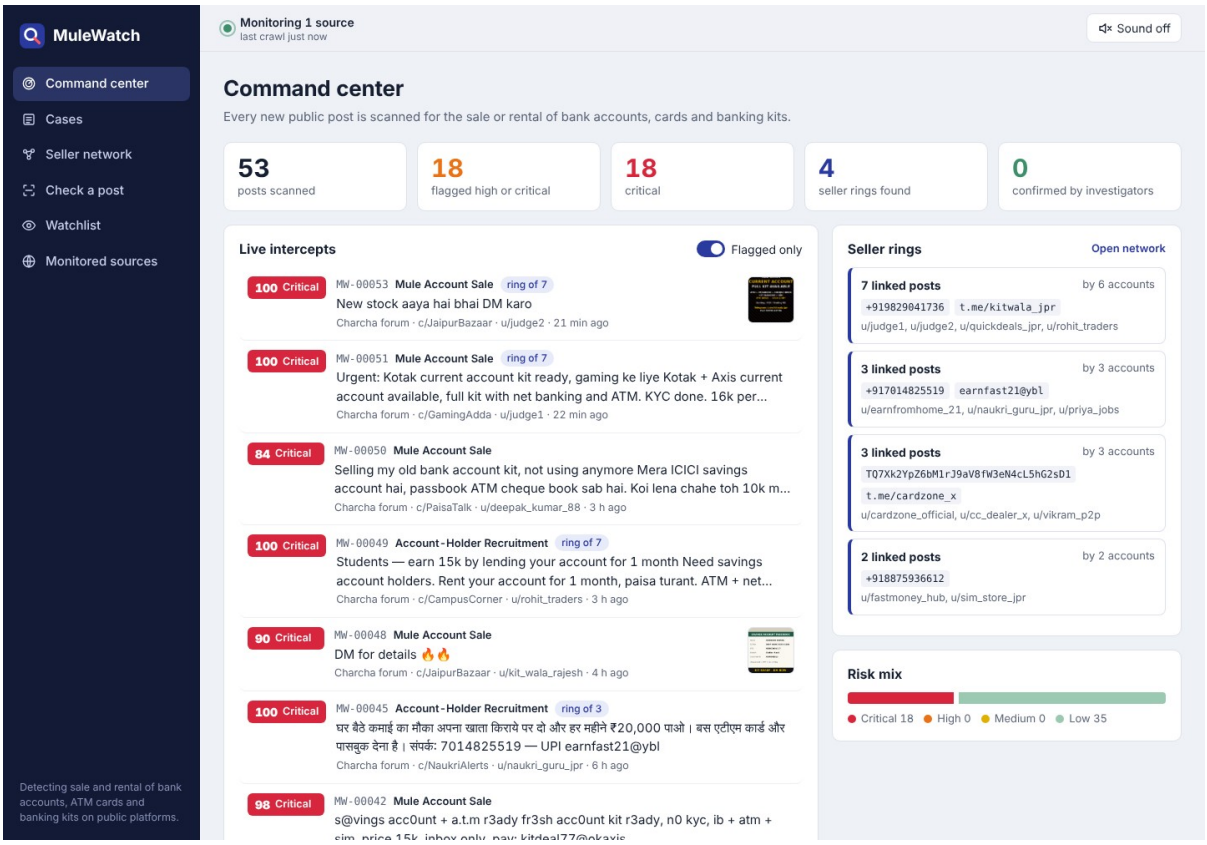
From your phone, post one test ad from section 6, watch the alert appear on the laptop, then reset the forum again (4.1) and restart with `./run.sh --reset` so the judges start from a clean 50-post state.

5. Demo script (about 5 minutes)

Timings are approximate. Each step lists what to do, a suggested line to say, and what the judges will see.

Step 1. Set the scene (30 s)

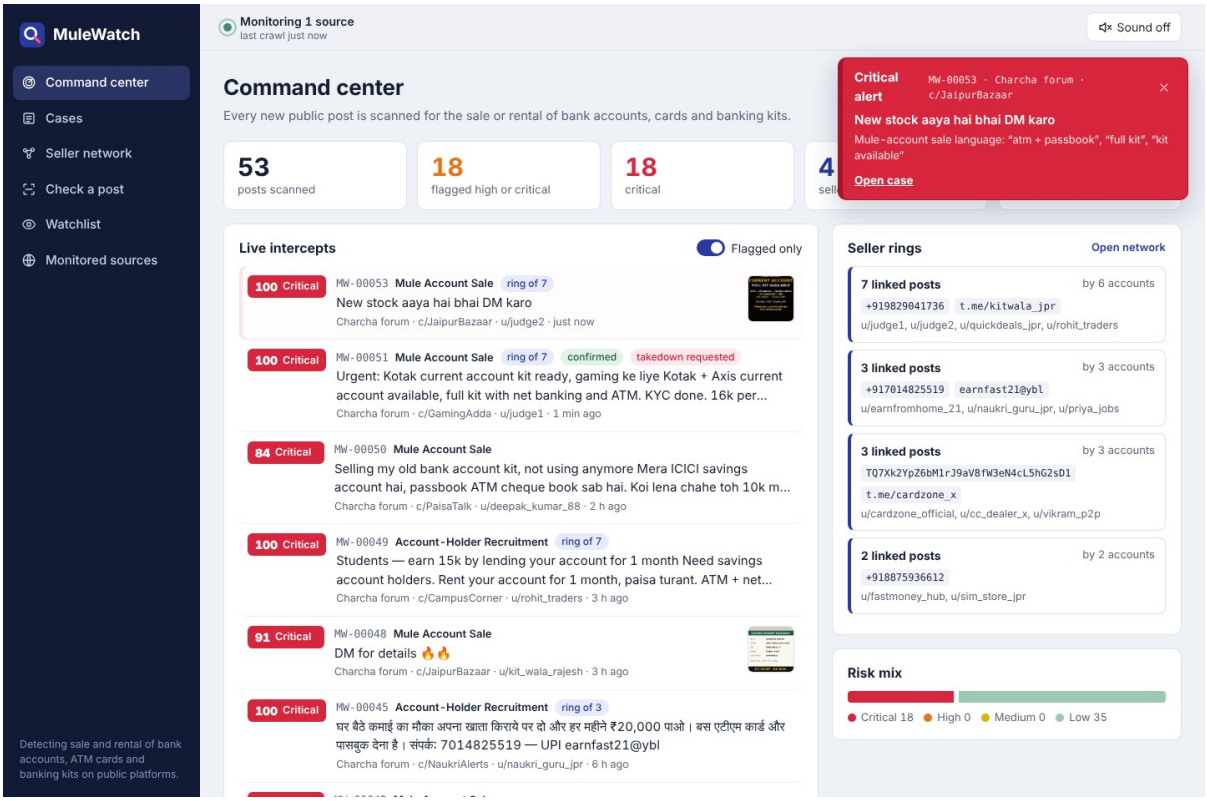
Do	Show the **Command center** at http://localhost:8000. Point at the numbers, the Live intercepts feed and the Seller rings panel.
Say	"Money-mule accounts are bought and rented openly on social media. MuleWatch watches public platforms in real time. It has already scanned 50 posts on this forum, flagged 16 as critical, and found 4 seller networks, without a single false alarm on the 34 normal posts."
Judges see	Dashboard with 50 scanned, 16 critical, 4 rings; red Critical rows in the live feed.



Command center: live intercepts, seller rings and risk mix

Step 2. A judge posts a mule ad live (60 s)

Do	Hand a judge a phone with http://103.155.85.18:8100 open and logged in as **judge1**. Ask them to tap **Create post**, choose **c/GamingAdda**, and paste the text of sample A from section 6, then tap **Post**.
Say	"Please post this like a seller would. Watch the screen."
Judges see	Within about 3 seconds a red **Critical alert** slides in at the top right with a chime, and the post appears at the top of the feed tagged **ring of 6**: it reused the Telegram handle of an existing seller network.



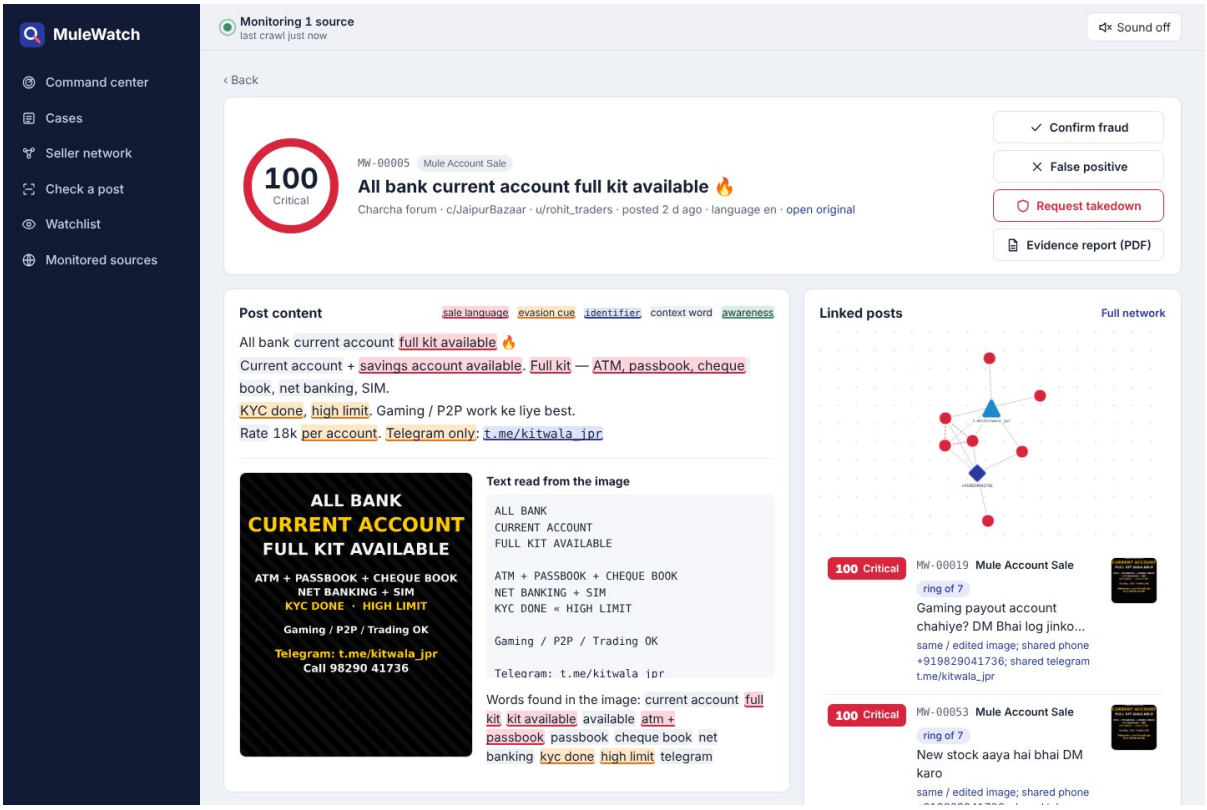
A Critical alert arriving in real time while the judge's post appears in the feed

Step 3. A harmless post stays quiet (30 s)

Do	Ask a second judge (judge2) to post sample B from section 6, an ordinary sale with a phone number.
Say	"Phone numbers and prices alone don't make something suspicious. The system looks for the combination of banking instruments, sale or rent intent, and evasion cues."
Judges see	The post shows up in the feed only when Flagged only is switched off, marked **Low**. No alert.

Step 4. Open the case and explain the evidence (60 s)

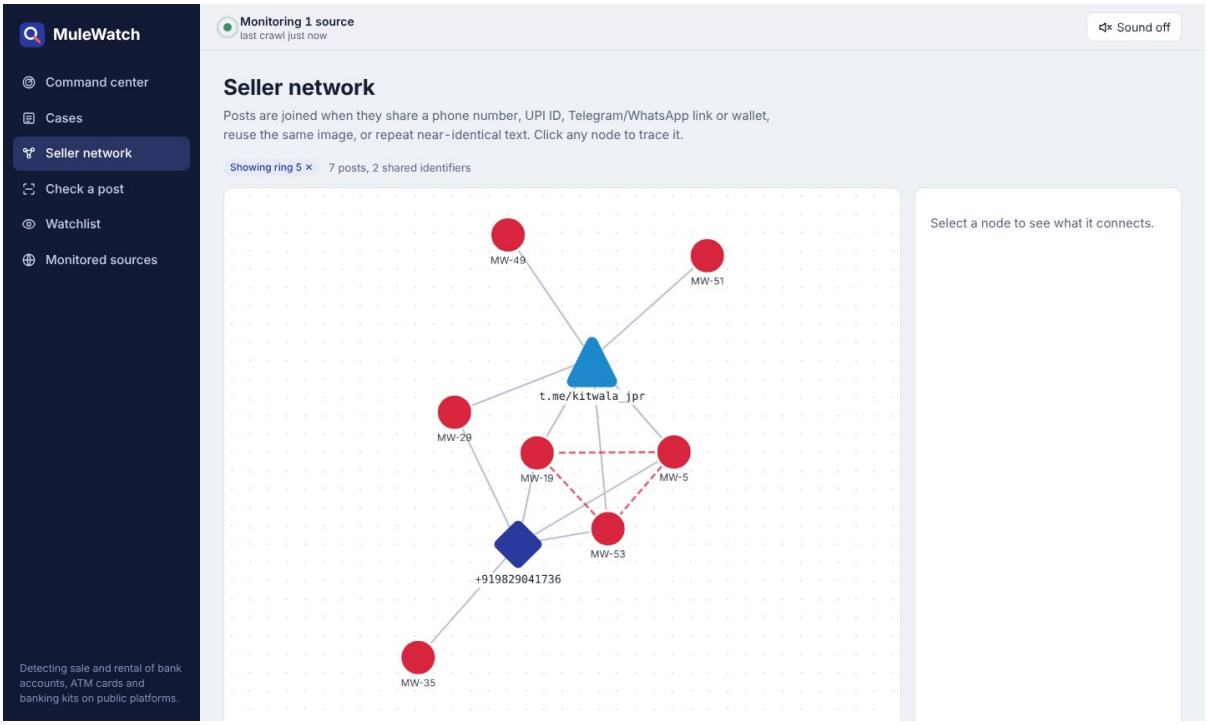
Do	Click **Open case** on the alert (or the post in the feed). Point at the highlighted words, the identifiers, **Text read from the image**, **Why it was flagged** and the score breakdown.
Say	"Every flag is explainable. Red is sale language, orange is evasion cues like KYC done, blue are identifiers. Here the image was read by OCR, so even image-only ads are caught. The score combines rules, a multilingual text model and image analysis."
Judges see	Score meter, colour-coded highlights, OCR text, reasons list and component scores. Linked posts on the right with why each one is linked.



Case view: highlighted evidence, OCR text from the poster, linked posts

Step 5. Show the seller network (45 s)

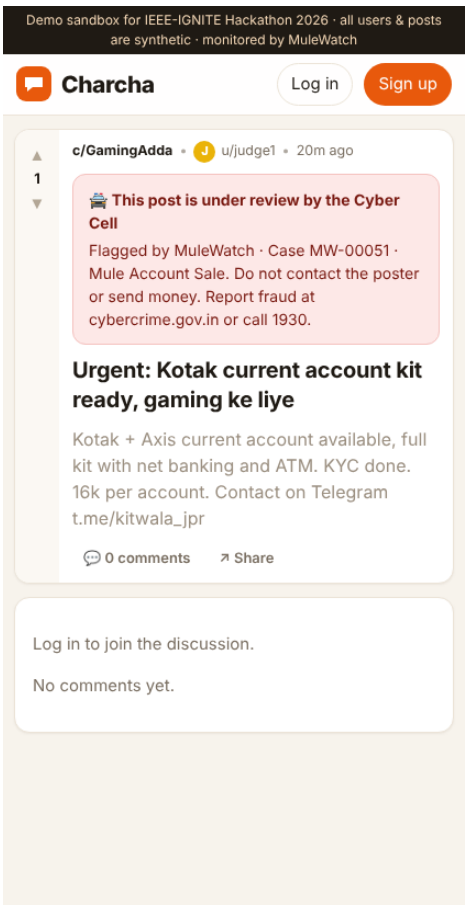
Do	Click **Full network** (or the **Seller network** menu). Click the Telegram triangle or phone diamond.
Say	"Six different accounts, one seller. They share a phone number and a Telegram handle, and two of them posted the same poster with a different crop and tint. MuleWatch links them automatically, so police can go after the network, not one post."
Judges see	Graph: red posts connected to shared identifiers; dashed red lines for reposted images or copied text. Clicking a node shows every post that used it.



One seller ring: posts linked through a shared phone number, Telegram handle and reposted poster

Step 6. Take action: takedown and evidence report (45 s)

Do	Back on the case, click **Request takedown**. Refresh the post on the judge's phone. Click **Evidence report (PDF)** and scroll through it. Optionally click **Confirm fraud**.
Say	"The investigator can ask the platform to take the post down; the forum immediately shows a Cyber Cell review notice. The PDF is ready to attach to an NCRP complaint, with a SHA-256 hash so the evidence can be shown to be unaltered."
Judges see	On the phone the post is greyed out with a red **under review by the Cyber Cell** banner. The PDF opens with case details, highlighted text, identifiers, linked posts and the hash.



The forum post after Request takedown

Step 7. Tricky cases and learning (30 s, if time allows)

Do	In **Cases**, search `Never sell` and open the PSA post; search `debit card` and open the student's post. On any borderline case, click **False positive**.
Say	"An awareness post that uses every keyword stays Low, and a student showing off a new debit card isn't a seller. When an investigator marks a mistake, the model retrains and rescores everything instantly."
Judges see	Both tricky posts are Low with an explanation. After False positive, a message confirms the model learned and the scores update.

6. Copy-paste posts for the live demo

Save these on the judges' phones (for example in a WhatsApp chat to yourself) so they can paste them quickly.

	Community	Title	Text	Expected result
A	c/GamingAdda	Urgent: Kotak current account kit ready	Kotak + Axis current account available, full kit with net banking and ATM. KYC done. 16k per account. Contact on Telegram t.me/kitwala_jpr	Critical, joins the kitwala ring
B	c/JaipurBazaar	Selling my cycle	Hero Sprint cycle, 1 year old, ₹3,000. Call 9876512345	Low, no alert
C	c/CampusCorner	Part time income for students	Apna bank account kiraye pe do, har mahine 25k milega. Sirf gaming transactions. WhatsApp 70148 25519	Critical, joins the rent-your-account ring
D	c/PaisaTalk	खाता उपलब्ध	करंट अकाउंट उपलब्ध, पूरा किट मिलेगा — एटीएम पासबुक चेकबुक। संपर्क 9876501234	Critical (Hindi)
E	c/JaipurBazaar	s@vings acc0unt r3ady	fr3sh acc0unt kit, a.t.m + ib, n0 kyc. inbox only	Critical (obfuscated spelling)
F	c/JaipurBazaar	New stock aaya hai	DM karo. Attach image kit_poster_repost.jpg (forum/seed/images on the laptop; copy it to the phone first)	Critical from the image alone; same-poster link
G	c/PaisaTalk	Beware of account rental scams	Never sell or rent your bank account. It makes you a money mule. Report to 1930.	Low (awareness)

> **Watchlist moment (optional)**
>
> Open **Watchlist**, add phone `9876512345` with note "tip from informant". The harmless cycle post (sample B) immediately becomes Critical with reason Watchlist hit. Remove it afterwards.

7. Backup plan and troubleshooting

7.1 If the server or venue internet fails

Run everything on the laptop. Judges on the same Wi-Fi can still post from their phones.

```
# stop the running command center with Ctrl+C, then:
cd ~/Projects/mulewatch
./run.sh --local-forum --reset

# give judges this address (laptop IP on the same Wi-Fi):
hostname -I           # e.g. 192.168.1.23  ->  http://192.168.1.23:8100
sudo ufw allow 8100/tcp   # only if Ubuntu's firewall is on
```

If phones can't reach the laptop (campus Wi-Fi often isolates devices), present from the laptop alone: open http://localhost:8100 in a second browser window, log in as judge1 and post from there while the judges watch the command center.

7.2 Common problems

Symptom	Cause	Fix
`./run.sh: No such file or directory`	Not inside the cloned folder, or an old folder with the same name	`cd ~/Projects/mulewatch` and check `ls run.sh`; if missing, move the folder aside and clone again (3.1)
`ensurepip is not available` or venv error	python3-venv not installed	`sudo apt install -y python3-venv`, then `rm -rf .venv` and run again
Top bar says Source unreachable	Laptop can't reach the forum	`curl -I http://103.155.85.18:8100`; check port 8100 in CyberPanel firewall; on server `systemctl status charcha`; or use 7.1
No alert sound	Browser blocked audio	Click Sound off so it becomes Sound on; unmute laptop
Image posts show no OCR text	Tesseract missing	`sudo apt install tesseract-ocr tesseract-ocr-hin`, restart run.sh
Request takedown fails	Token mismatch	Same `FORUM_MOD_TOKEN` in laptop `.env` and server service; re-run setup script with the token
Posts scanned higher than 50 before demo	Test posts from rehearsal	Reset forum (4.1) and `./run.sh --reset`
Command center opened on 8001, not 8000	Another program on the laptop already uses port 8000; run.sh picks the next free port	Use the address run.sh prints, or choose one: `./run.sh --port 8010`
Old MuleWatch still running in another terminal	A previous run.sh was not stopped	Press Ctrl+C in that terminal, or `fuser -k 8000/tcp 8100/tcp`
Request takedown says the post is no longer on the forum	The forum was reset after this case was captured	Expected; take down a post that exists on the forum now

Symptom	Cause	Fix
Forum page error after server reboot	Service stopped	`sudo systemctl restart charcha`; logs: `journalctl -u charcha -n 50`

8. Likely questions from judges

How accurate is it?

On the 50 forum posts, which the model never saw during training (16 mule or card ads, 34 normal or tricky), it caught all 16 with no false alarms. Run `python training/evaluate.py` in `backend/` to reproduce. It is a small synthetic set, so we treat it as a sanity check; the feedback loop is how accuracy improves with real cases.

Is this just keyword matching?

No. Rules are one of three signals. A character n-gram model handles Hinglish spelling variants and obfuscation; OCR covers images; ring detection adds network evidence. Awareness posts that contain every keyword stay Low because context lowers risk.

How would it work on real platforms?

The collectors are pluggable. The forum crawler reads standard schema.org structured data that most forums publish; the Telegram collector reads public channel previews without login; the Chrome extension lets an investigator send anything they see. Platform APIs or trust-and-safety feeds can be added as new sources.

Why character n-grams instead of a large model?

It runs offline on a laptop in milliseconds, retrains instantly from investigator feedback, and handles Hindi, Hinglish and leetspeak without a tokenizer. An optional Claude second opinion can be switched on with an API key.

What about privacy and misuse?

Only publicly available content is collected. Card numbers are masked. Scores are leads for a human investigator, not automatic punishment; every case has a verdict step and an evidence trail.

How does it detect coordination?

Three independent links: near-identical text (MinHash), the same image even if cropped or recoloured (perceptual hash), and shared identifiers such as phone, UPI, Telegram or wallet. Connected posts form a ring, and ring membership raises the score.

Can it scale?

Each post is analysed in milliseconds. SQLite and a single process are enough for the demo; the same code runs on PostgreSQL with a task queue for production volumes.

9. Command reference

Task	Command	Where
Start command center	`./run.sh`	Laptop, in ~/Projects/mulewatch
Start on a specific port	`./run.sh --port 8010`	Laptop
Start with clean cases	`./run.sh --reset`	Laptop
Start with local backup forum	`./run.sh --local-forum`	Laptop
Update code	`git pull`	Laptop
Run tests	`cd backend && ../.venv/bin/python -m pytest -q`	Laptop
Accuracy on forum posts	`cd backend && ../.venv/bin/python training/evaluate.py`	Laptop
Reseed local forum	`cd forum && ../.venv/bin/python seed_forum.py`	Laptop
Copy forum to server	`scp -r mulewatch/forum mulewatch/deploy root@103.155.85.18:/opt/charcha-src/`	Laptop
Install / update forum	`sudo FORUM_MOD_TOKEN=skit-cybercell-2026 bash /opt/charcha-src/deploy/setup-forum-server.sh`	Server
Reset forum to 50 posts	`sudo RESEED=1 FORUM_MOD_TOKEN=skit-cybercell-2026 bash /opt/charcha-src/deploy/setup-forum-server.sh`	Server
Forum status / restart	`systemctl status charcha` / `sudo systemctl restart charcha`	Server
Forum logs	`journalctl -u charcha -f`	Server
Check forum from laptop	`curl -I http://103.155.85.18:8100`	Laptop
Free busy ports	`fuser -k 8000/tcp 8100/tcp`	Laptop

All users, phone numbers, handles and banks in the demo data are fictional. The forum is marked as a sandbox and blocks search engines.